A Wall-Climbing Air-Coupled GPR Robot and a Lightweight Signal Network for Automated Tunnel Lining Inspection

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***Abstract*—Concrete lining quality is critical to the safety of railway tunnels, yet internal defects such as voids, insufficient compaction, and inadequate thickness are still inspected largely by hand-held ground-penetrating radar (GPR) operated from lift trucks or scaffolding. This practice is slow, unsafe, disrupts construction, and produces paper reports that are hard to trace. We present an end-to-end inspection system that combines (i) an air–ground amphibious wall-climbing robot carrying a 900 MHz wireless air-coupled GPR, (ii) a lightweight one-dimensional convolutional network that classifies raw radar signals into four defect-related classes and regresses the depth range of detected voids, and (iii) a cloud management platform that maps every defect onto a 3D tunnel model, grades its severity, and drives a closed remediation loop. We build a dual-modality dataset of radargram images and raw signals collected by the robot in two railway tunnels. On a balanced set of 4,000 labeled signals, the signal network reaches a macro-F1 of 0.93 while reducing computational complexity by 65%, and it generalizes better than a YOLOv5 detector trained on radargram images. The system has been deployed on several railway projects, with more than 160 km of lining inspected.**

***Index Terms*—Ground-penetrating radar, tunnel inspection, wall-climbing robot, 1D convolutional neural network, YOLO, digital twin, structural health monitoring.**

I. Introduction

China's railway tunnel inventory has expanded rapidly (Fig. 1). In 1949 the network contained 429 tunnels with a combined length of 112 km. By the end of 2024, the railway network had reached an operating length of 159,000 km, including 16,798 operating tunnels with a total length of about 19,630 km. Managing construction quality at this scale is a demanding task with direct consequences for national development and public safety.

The concrete lining is the structural "lifeline" of a tunnel. Complex geology, long construction spans, construction technique, site conditions, and variable site management make linings prone to internal defects, including voids behind or within the lining, insufficient compaction, and inadequate thickness. Such defects are invisible from the surface, can grow over time, and lead to costly repair or reinforcement if not found early. Lining quality inspection is therefore one of the main tools for safeguarding tunnel construction.

Current practice relies on manual GPR surveys, which suffer from three groups of problems.

Unsteady, slow acquisition. Operators hold the antenna against the lining while standing on a lift truck. The GPR unit consists of a host, a cable, and an antenna, and the cable is only about 15 m long, so operators must carry the host and walk with the truck. Because the truck sways while moving and the hand-held antenna is unsteady, data along a survey line are often unreliable and must be re-collected.

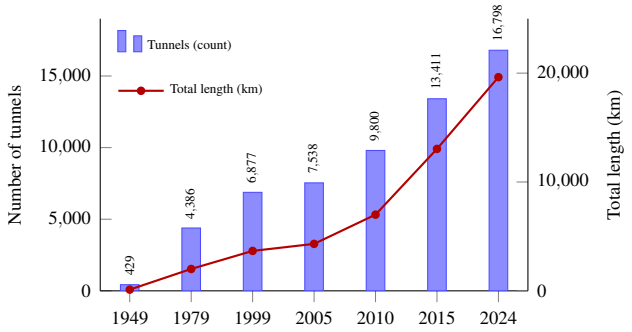


Figure 1. Growth of China's railway tunnel network: number of tunnels (bars) and total length (line).

Disruption and cost. The alternative is a mobile scaffold built around the tunnel cross-section, which lets construction vehicles pass underneath but requires 12–14 workers to push over uneven ground and consumes considerable material and time. Dedicated inspection vehicles are expensive and hard to move along mountain roads, so full coverage of every tunnel is rarely achieved.

Slow, opaque reporting. Results are delivered as lengthy paper reports in which defect locations are hard to find, compare across projects, or trace back to raw data. As a result, site managers lack timely knowledge of lining quality, and serious problems may not be escalated to responsible staff in time.

This paper describes an integrated "equipment–data–analysis" pipeline that automates the full inspection loop, from data acquisition on the lining to remediation tracking in the office. Our contributions are:

- an air–ground wall-climbing robot that carries a wireless air-coupled GPR along the curved lining at constant speed, switching between flight and climbing modes using onboard pressure sensing (Section III);
- a cloud management platform that turns inspection results into a 3D, traceable, closed-loop quality record with graded alerts (Section IV);
- a lightweight 1D-CNN that operates directly on raw GPR

Table I
REPRESENTATIVE TUNNEL INSPECTION SYSTEMS

System	Origin	Sensing	Speed (km/h)	Crack res. (mm)	Target
TS3	DE	Laser + IR	5	0.2	Highway
GRP 5000	CH	Laser + vision	2–4	0.3	Railway
tCrack	CH	Vision	2.5	0.3	Railway
MTI-100	CN	Vision	5	0.3	Metro
Huoyanhou	CN	Vision	30–40	0.2	Metro
MIMM-R	JP	Radar + vision + laser	80	0.2	Highway
Lining insp. vehicle	JP	Vision + hammering	0.6	0.5	Highway
ZOYON-TFS	CN	Vision + IR + laser	80	0.2	Highway

traces, jointly classifying four signal types and estimating void depth through a regression head (Section V);

- a dual-modality dataset of radargram images and raw signals collected with the robot (Section VI), a comparison against an image-based YOLOv5 detector (Section VII), and field deployment over more than 160 km of lining (Section VIII).

II. RELATED WORK

A. Tunnel Inspection Systems

Table I summarizes representative tunnel inspection systems developed in Europe, Japan, and China. Most rely on machine vision, laser scanning, or infrared imaging, and their performance is usually characterized by survey speed and crack resolution. Such systems are effective for surface distress such as cracks, but few of them can see defects hidden inside or behind the lining.

Many of these devices are still at the research or trial stage. Comparing them reveals three recurring gaps.

- *Low automation.* Data can be acquired automatically, but skilled operators still have to run and follow the equipment. Reducing risk and operator subjectivity while improving speed and accuracy requires fully automated systems that draw on AI, big data, and robotics.
- *Low functional integration.* Most platforms are designed for a single environment (highway or railway) and carry a single sensor. The Lynx system, for example, can only detect voids behind the lining, whereas surface distress, internal damage, and the state of the surrounding rock all matter during operation.
- *Low intelligence.* In-tunnel positioning, obstacle avoidance, storage, recognition, transmission, analysis, and assessment are rarely automated. Some devices use machine learning to recognize and quantify defects, but the results still need manual confirmation.

B. Ground-Penetrating Radar for Lining Inspection

GPR transmits short electromagnetic pulses into the structure and records the reflections from interfaces between media with different dielectric properties [1, 2]. In a medium with relative permittivity ε_r, the wave travels at approximately

$$v = \frac{c}{\sqrt{\varepsilon_r}}, \tag{1}$$

where c is the speed of light in vacuum. At normal incidence on an interface between media 1 and 2, the amplitude reflection coefficient is

$$\Gamma = \frac{\sqrt{\varepsilon_{r1}} - \sqrt{\varepsilon_{r2}}}{\sqrt{\varepsilon_{r1}} + \sqrt{\varepsilon_{r2}}}. \tag{2}$$

Air-filled voids and poorly compacted concrete differ strongly in permittivity from sound concrete, which produces characteristic echoes. Rebar produces strong point-like reflections that appear as hyperbolas in a radargram and can mask weaker defect echoes. By measuring the two-way travel time of an echo, the depth of the reflecting interface follows from (1). Lining thickness, voids, and regions of insufficient compaction can therefore all be assessed from GPR data.

Conventional ground-coupled antennas must be pressed against the surface. On a tunnel lining this means a person, a lift truck, or a scaffold must hold the antenna in contact with the concrete along the full survey line. Air-coupled antennas instead operate at a standoff from the surface, which is what makes robotic acquisition on a curved lining practical.

C. Deep Learning for GPR Interpretation

Interpreting radargrams manually is slow and depends on expert experience, which has motivated automatic interpretation with deep learning. The most common approach treats the radargram as an image and applies object detectors. Prior studies report that one-stage YOLO detectors [3, 4] outperform two-stage Faster R-CNN [5] on radargrams. Image-based detection, however, needs many annotated radargrams, and valid defect samples are scarce because dense rebar reflections dominate lining radargrams.

An alternative is to process each A-scan (single trace) directly with a one-dimensional convolutional network. 1D CNNs are compact, learn features directly from raw time series, and have been applied widely to signal classification tasks [6]. Working on traces also allows the network to read echo timing directly, which is what quantitative void measurement requires. Our method follows this second approach.

III. WALL-CLIMBING GPR ROBOT

A. Design Requirements

The pain points above translate into five requirements for the acquisition platform: (1) no cable tether between antenna and host; (2) stable antenna standoff and constant speed along a survey line, so that each line is captured once; (3) the ability to reach the crown and follow the curved lining circumferentially; (4) no obstruction of construction traffic; and (5) real-time delivery of data to an analysis platform.

B. Mechanical Design and Locomotion

The robot (Fig. 2) uses a quadrotor plus four-wheel-drive design. We set mechanical performance targets so that the robot can travel circumferentially along the curved lining surface. Its architecture comprises a radar subsystem, a sensing subsystem, and a carrying subsystem.

Carrying subsystem. By regulating the power output of each rotor, the robot carries the radar to the tunnel crown or hovers against the wall. Once it reaches the start of a survey line, its bottom wheel set drives the radar along the line at constant



(a) Tiehang-1 (b) Tiehang-2

Figure 2. The two wall-climbing GPR robot prototypes. The radar is mounted beneath the chassis; rotors provide thrust for flight and wall adhesion, and the wheels drive the robot along the lining.

speed while rotor thrust keeps the wheels pressed against the lining.

Sensing subsystem. Pressure sensors at each wheel support measure the contact force between the robot and the lining, and changes in this force trigger automatic switching between flight mode and climbing mode. A gravity sensor provides an orientation reference used to correct the survey path.

Two prototypes were built: a quadrotor with a wheeled undercarriage (Tiehang-1) and a caged four-wheel climber with ducted rotors (Tiehang-2).

C. Radar Subsystem

A TH-900 air-coupled wireless GPR is mounted under the chassis. We developed a 900 MHz remotely controlled wireless GPR host from mature components, integrating the control unit and a GC900HF antenna into a single module, which removes the cable tether entirely. The radar subsystem consists of three modules:

- a *timing-control module* that sets the acquisition rate from the rotation count of a distance-measuring wheel, so traces are spaced uniformly along the survey line;
- a *signal-acquisition module* with transmit and receive antennas that excite the radar wave and record its echoes after propagation through the lining;
- a *wireless module* that streams the acquired traces over Wi-Fi to a ground computer.

The electromagnetic propagation obeys Maxwell's equations, so the state of the medium can be inferred from the received echoes. By tuning signal parameters and processing algorithms, the system compensates for the effect of the air gap on propagation and on the transmitted signal within a certain distance. The robot can therefore survey at a standoff from the lining instead of pressing the antenna against it, which avoids obstacle conflicts during contact surveys and keeps each survey line continuous. The radar was designed around three core capabilities: a high-dynamic-range, high-precision receiver; Wi-Fi data communication; and low power consumption for airborne operation.

D. Operating Procedure

A survey proceeds as follows. The operator selects a survey line from the inspection plan. The robot takes off in flight mode and flies to the start of the line, at the crown or on the side wall. When the wheel pressure sensors register contact, it switches to climbing mode and drives along the line at constant speed, while the distance wheel triggers trace acquisition and the traces stream wirelessly to the ground station. At the end of the line, the robot detaches and flies to the next line or lands. Control combines flight control and vehicle control over a local wireless network. Because the robot works on the lining surface, construction vehicles below are not obstructed.

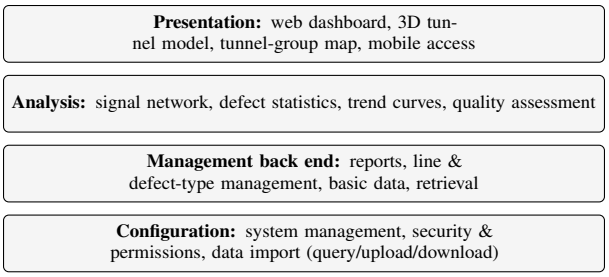


Figure 3. Layered architecture of the tunnel lining quality management platform.

IV. Intelligent Management Platform

Efficient acquisition is only half of the problem; results must also be processed, visualized, and acted on quickly. We therefore developed a cloud-based concrete quality monitoring platform that fuses and shares inspection information. It visualizes lining quality, assesses and predicts quality trends, links data to anticipate safety risks, pushes feedback to responsible staff, and supports tracing problems back to their source.

A. Architecture

The platform has four layers (Fig. 3). The *configuration layer* provides system management and security, together with import management for field data, including report query, upload, and download. The *management back end* organizes and analyzes imported data through inspection-report management, statistical analysis, and basic-data management, and supports retrieval and compliance comparison. The *analysis layer* runs defect recognition and big-data analytics on the stored records. The *presentation layer* is a web dashboard that renders the overall inspection status of every tunnel, with access from multiple terminals including mobile devices.

B. 3D Visualization and Defect Encoding

Field data are packaged and imported into the back end in one step. Using station (chainage) numbers as the link, the platform places each record on a 3D digital model of the tunnel, rendering defect size, standard survey lines, and lining joints to scale. Users can switch between front, left, and right views. A "tunnel-group map" gives an overview of all tunnels in a project, built from tunnel metadata, inspection data, and repair records. Clicking a defect's station number jumps to its location in the 3D model and shows the detection result, radar image, inspection table, and report. Defect types are color-coded as in Table II, and defects are graded into severity levels by numeric intervals.

Table II
DEFECT COLOR ENCODING IN THE 3D TUNNEL MODEL

Color	Meaning
Red	Insufficient lining thickness
Blue	Void or insufficient compaction
Orange	Circumferential rebar problem
Gray	Anti-crack steel mesh problem
Green	Defect remediated and re-inspection passed

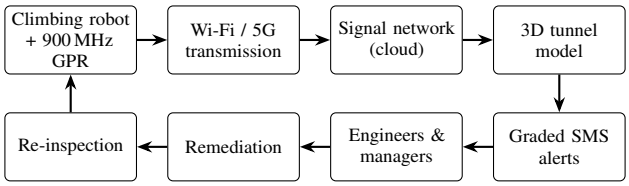


Figure 4. End-to-end inspection pipeline, from data acquisition to closed-loop remediation.

C. Analytics and Alerts

The statistics module aggregates defects by contract section, division, and tunnel. The inspection-management module analyzes data from several angles: inspection reports, repair reports, survey-line management, defect-type management, and verification videos. Keyword search and new-report import support fast queries. On the project page, defect categories are compared and projects are ranked by defect rate, so recurring problems in a contractor's work become visible. For each tunnel, the platform plots how defects develop over time, giving a picture of the crew's management, technique, and process trends and generating reports that let the contractor adjust its process. Severity grades drive SMS alerts to the responsible company and project staff. Role-based permissions, assigned by identity and position, protect the database.

D. Closed-Loop Remediation

Each defect record links its radar image, verification video, remediation plan, defect-removal and grouting footage, and re-inspection image. This enforces a chain of "one problem, one verification, one plan, one remediation, one closure." When a repaired defect passes re-inspection, its color in the model changes to green and its status changes from "not remediated" to "remediated," with the re-inspection radar image attached. Fig. 4 summarizes the full loop. Replacing paper reports with this digital record also removes printing and storage costs and makes every result accessible from any terminal.

V. LIGHTWEIGHT RADAR SIGNAL NETWORK

Rather than detecting defects in radargram images, we process each raw GPR trace with a lightweight network made of two 1D-CNN branches on a shared encoder: one classifies the signal and the other measures the extent of voids (Fig. 5). The network consists of an input layer, 1D convolutional layers, a 1D classification layer, a 1D regression layer, and an output layer.

Table III
SIGNAL CLASSES AND ONE-HOT LABEL VECTORS

Class	Signal type	Label vector
ω_1	Rebar	$[1,0,0,0]$
ω_2	Insufficient thickness	$[0,1,0,0]$
ω_3	Insufficient compaction	$[0,0,1,0]$
ω_4	Void	$[0,0,0,1]$

A. Problem Formulation

Let $x \in \mathbb{R}^L$ be a single GPR trace of L samples, i.e., the electric field strength recorded over the two-way travel-time window at one station. Each trace belongs to one of four classes, ω_1 rebar, ω_2 insufficient thickness, ω_3 insufficient compaction, and ω_4 void, encoded as one-hot vectors (Table III). For void traces we additionally seek the two-way travel times t_{top} and t_{bot} of the reflections from the top and bottom boundaries of the void.

B. Feature Extraction

The encoder maps x to a feature vector F through stacked 1D convolutional layers. Layer l computes

$$h_c^{(l)}[n] = \sigma\Big(b_c^{(l)} + \sum_{c'} \sum_{k=0}^{K-1} w_{c,c',k}^{(l)} h_{c'}^{(l-1)}[sn+k]\Big), \qquad (3)$$

where K is the kernel size, s the stride, c and c' index output and input channels, and σ is a nonlinear activation. Kernel size and stride are adjustable. Multiple kernels slide over the signal to extract features efficiently. Shallow layers capture physical attributes such as field strength, phase, and travel time, while deeper layers iteratively learn semantic features of the target echoes. A fusion layer combines features across levels into the composite vector F.

C. Classification Head

A fully connected layer maps F to logits $z \in \mathbb{R}^4$, and a softmax produces the class distribution

$$p_i = \frac{\exp(z_i)}{\sum_{j=1}^{4} \exp(z_j)}, \qquad i = 1,\ldots,4. \qquad (4)$$

The predicted class is $\hat{\omega} = \arg\max_i p_i$. When p_4, the void probability, is the largest, the void measurement branch is triggered.

D. Void Depth Regression

For a trace classified as ω_4, the features F that characterize the reflections from the top and bottom of the void are passed to the 1D regression layer, which predicts the two time windows. The raw time-window values are large, and regressing them directly makes convergence difficult, so targets are first normalized to $[0,1]$ by the length T of the recording window,

$$\tilde{t} = t/T, \qquad (5)$$

and the regression layer uses a sigmoid activation, whose range matches the normalized targets. Denormalizing the outputs

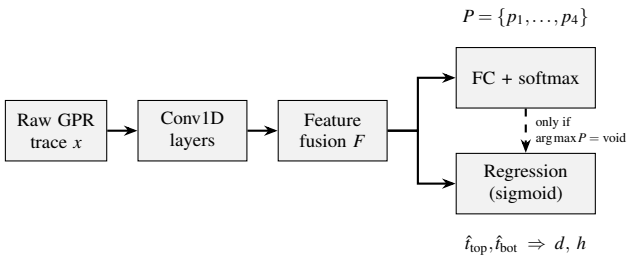


Figure 5. Lightweight radar signal network: a shared 1D convolutional encoder feeds a four-class classifier and, for void signals, a regression head that estimates the echo times of the void boundaries.

gives $\hat{t}_{\text{top}}$ and $\hat{t}_{\text{bot}}$. With the wave velocity v from (1), the depth of the void top and the vertical extent of the void are

$$d = \frac{v\,\hat{t}_{\text{top}}}{2}, \qquad h = \frac{v\,(\hat{t}_{\text{bot}} - \hat{t}_{\text{top}})}{2}. \tag{6}$$

E. Training Objective

The classification head is trained with multi-class cross-entropy,

$$\mathscr{L}_{\text{cls}} = -\sum_{i=1}^{4} y_i \log p_i, \tag{7}$$

where y is the one-hot label. The regression head is trained with mean squared error on void samples,

$$\mathscr{L}_{\text{reg}} = \tfrac{1}{2}\Big[(\tilde{t}_{\text{top}} - \hat{\tilde{t}}_{\text{top}})^2 + (\tilde{t}_{\text{bot}} - \hat{\tilde{t}}_{\text{bot}})^2\Big]. \tag{8}$$

F. Lightweight Design

The architecture maps a raw signal to a defect class and a void measurement end to end in a single pass. Through its lightweight design, it reduces computational complexity by 65% while maintaining detection accuracy, which meets real-time requirements on site.

VI. Dataset Construction

A. Data Acquisition

Using the robot, we collected 5 km of lining data in the Xiaolijian and Lushan tunnels of the Huzhou–Hangzhou Railway. From these data we built a dual-modality dataset: an image dataset of radargrams and a time-series dataset of raw radar signals (Table IV). Both are balanced across the four classes.

B. Raw Signal Dataset

Fig. 6 shows the labeling workflow. The TH-900 radar records raw data in `.lte` format. We converted the files to `.rd3` and processed them in Reflexw [7] to export XYZ feature files with three columns: station coordinate, time window (depth), and electric field strength. We created a dedicated document for each of the four classes following a predefined coding scheme. Annotators marked the station of each target signal in Reflexw, computed the corresponding survey line from the total number of lines and the station, extracted the field-strength trace for that line, and stored it as a CSV file under its class. For void signals, annotators additionally traced the upper and lower edges of the reflection hyperbola to fix the depth time window. This yields dual labels holding the top and bottom echo times.

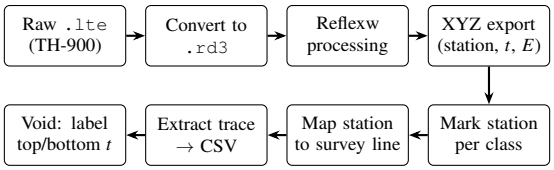


Figure 6. Labeling workflow for the raw signal dataset.

Table IV
Datasets Collected with the Robot

	Radargram images	Raw signals
Source	5 km, Xiaolijian & Lushan tunnels	
Classes	4	4
Samples/class	300	1,000
Total	1,200	4,000
Extra labels	bounding boxes	void top/bottom times
Augmentation	crop, mirror	–
Annotation	LabelImg, two-stage	Reflexw, manual
Train / val	8:2	3,200 / 800

The final dataset contains 4,000 anomalous traces, 1,000 per class, so voids make up 25% of the data. It was split 8:2 into 3,200 training and 800 validation traces, which satisfies the distribution requirements of deep network training.

C. Radargram Image Dataset

Radargram recognition models face a data scarcity problem: rebar layers interfere with defect echoes, so valid samples are rare. We applied sliding-window cropping and mirror flipping as data augmentation, obtaining 1,200 anomalous radargrams, 300 per class. Images were annotated with LabelImg [8] and split 8:2.

To reduce annotation cost, the image detector was trained in two stages. An initial model was first trained on a small manually labeled subset and used to pre-annotate the remaining images, which were then used for training. This lowered labeling effort, improved label consistency, and reduced the expertise needed to use and retrain the model.

VII. Experiments

A. Setup

All models were implemented in PyTorch [9]. Training ran on an Intel Core i5-7500 CPU (3.40 GHz) with an NVIDIA GeForce GTX 1080 Ti GPU. Table V lists the signal network's training configuration. Because prior studies report that one-stage YOLO models outperform two-stage Faster R-CNN on radargrams, we compare only the signal network with YOLOv5s [4] trained on the image dataset.

B. Evaluation Metrics

Since the task has four classes and the classes are balanced, we use macro-averaging to evaluate classification performance

Table V
TRAINING CONFIGURATION OF THE SIGNAL NETWORK

Item	Setting
Framework	PyTorch
Hardware	Intel i5-7500, NVIDIA GTX 1080 Ti
Optimizer	NAdam [10]
Learning rate	1×10^{-3}
(β_1, β_2)	(0.9, 0.999)
Epochs	200
Batch size	5 signals
Classification loss	Multi-class cross-entropy
Regression loss	Mean squared error

over all classes. For class i,

$$P_i = \frac{TP_i}{TP_i + FP_i}, \qquad R_i = \frac{TP_i}{TP_i + FN_i}, \qquad F1_i = \frac{2P_iR_i}{P_i + R_i}, \tag{9}$$

and the macro-averaged scores over $k = 4$ classes are

$$P_{\text{macro}} = \frac{1}{k}\sum_{i=1}^{k} P_i, \quad R_{\text{macro}} = \frac{1}{k}\sum_{i=1}^{k} R_i, \quad F1_{\text{macro}} = \frac{1}{k}\sum_{i=1}^{k} F1_i. \tag{10}$$

Average precision is the area under the precision–recall curve, and mAP averages it over n classes [11]:

$$AP = \int_0^1 P(R)\,dR, \qquad mAP = \frac{1}{n}\sum_{i=1}^{n} AP_i. \tag{11}$$

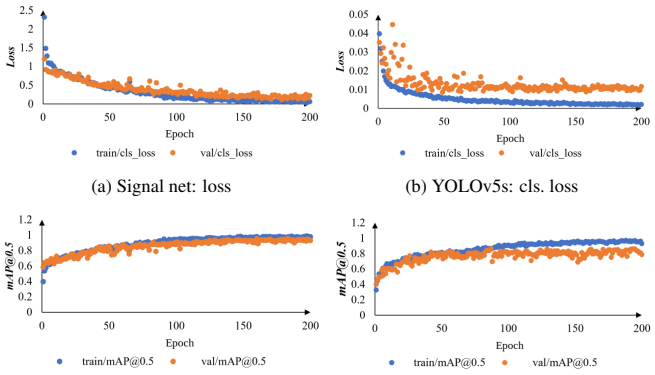
We report mAP@0.5 during training. Model complexity is measured by floating-point operations (GFLOPs, $1\,\text{GFLOPs} = 10^9$ FLOPs), where more FLOPs indicate a more complex model, and by the number of trainable parameters, where more parameters indicate a more complex feature extractor.

C. Results

The signal network reaches a macro-F1 of **0.93** on the validation set. Fig. 7 compares training dynamics. For the signal network, training and validation losses decrease together, and validation mAP@0.5 rises steadily to about 0.9, tracking the training curve closely throughout. YOLOv5s fits the training images well, with training mAP@0.5 near 0.95. However, its validation mAP@0.5 plateaus around 0.8–0.85 after roughly 50 epochs, and its validation classification loss stays well above the training loss for the rest of training.

D. Discussion

We see four reasons why the signal-based approach generalizes better on this problem. First, *data efficiency*: each radargram yields many traces, so the same 5 km of survey data supports 4,000 balanced signal samples but only 1,200 images, even after augmentation. The persistent train–validation gap of YOLOv5s is consistent with overfitting a large detector to a small image set. Second, *interference*: in images, strong rebar hyperbolas visually dominate weaker defect echoes, whereas the signal network learns from labeled traces of each class directly. Third, *measurement*: a trace preserves echo timing explicitly, which allows the regression head to estimate void depth and height via (6), a quantity that bounding boxes on a radargram do not provide directly. Fourth, *cost*: 1D convolutions over a single trace are far cheaper than 2D convolutions over an image, which suits on-site and cloud deployment at scale.



(a) Signal net: loss (b) YOLOv5s: cls. loss

(c) Signal net: mAP@0.5 (d) YOLOv5s: mAP@0.5

Figure 7. Training (blue) and validation (orange) curves over 200 epochs. The signal network generalizes well, while YOLOv5s shows a persistent train–validation gap.



Figure 8. Field inspection: the climbing robot surveying the lining of a railway tunnel.

The image-based detector still has a role. It gives inspectors a familiar view of the radargram and served as the first recognition model in the system, and the two-stage labeling procedure keeps its annotation cost low.

VIII. FIELD DEPLOYMENT

A. Validation

The robot was first deployed in the Xiaolijian and Lushan tunnels of the Huzhou–Hangzhou Railway, where detected defects were confirmed by on-site borehole drilling (Fig. 8). The engineering application and verification results show that the robot acquires and detects lining damage automatically with good stability and accuracy. Further field tests on the Xiongxin project (Taiyuan section) and the Jining–Datong–Yuanping Railway verified the robot's stability in complex tunnel environments, including borehole verification and data calibration.

B. Large-Scale Application

The system has since been used on the Chongqing–Qianjiang, Hangzhou–Wenzhou, and Quzhou–Lishui railway projects,

Table VI
ROBOTIC VS. CONVENTIONAL LINING INSPECTION

Aspect	Conventional (hand-held GPR)	Robot + platform
Acquisition	Operator holds heavy antenna; shaking degrades data	Antenna held at stable standoff by the robot
Efficiency	Speed depends on operator, fatigue, and frequent repositioning	Constant speed along preset lines, little intervention
Environment	Hindered by narrow space and site equipment; may delay construction	Works on the lining without blocking construction
Data	Manual collection; slow transfer, risk of loss; delayed analysis	Wireless transfer with redundant storage; 3D view; automatic alerts
Safety	Work at height, risk of falls and falling objects	No personnel at height

among others, with a cumulative inspected length of more than 160 km across railway, metro, and hydraulic tunnels. Through cooperation with construction enterprises, inspection data are linked directly with construction management on these projects.

C. Comparison with Conventional Practice

Table VI contrasts robotic inspection with the conventional hand-held approach along five dimensions observed in deployment.

IX. LIMITATIONS

Our study has several limitations. The training data come from two tunnels on one railway line, so performance on other lining designs, geology, and radar settings remains to be measured. We report aggregate rather than per-class metrics and do not yet report the accuracy of the void depth regression against borehole measurements; both are needed for a complete evaluation. The comparison with YOLOv5s uses different input modalities and sample counts, so it reflects the full pipeline rather than the model architectures alone. Finally, the robot still relies on an operator for mission planning and does not yet avoid obstacles autonomously.

X. FUTURE WORK

A. Recognition Algorithms

Generative adversarial networks [12] could synthesize realistic samples of rare defect patterns, such as unusual shapes, sizes, and positions or cluttered backgrounds, to enrich training data where field samples are scarce. Reinforcement learning [13] could let the recognition pipeline adapt detection parameters and processing modules to site conditions, rewarding choices that improve accuracy and efficiency.

B. Multi-Source Data Fusion

GPR reveals internal defects but says little about overall geometry or surface state. We plan to fuse GPR with laser scanning, which provides an accurate 3D frame for locating and sizing defects, and with infrared thermography, which detects surface temperature anomalies caused by internal defects. 5G and IoT links would carry these heavier data streams to the processing center quickly and reliably.

C. New Sensors

Candidate sensors include miniaturized, high-sensitivity MEMS and millimeter-wave radar, non-contact microwave inspection for fast large-area screening, and ultrasonic phased arrays for fine characterization of suspected defects. Combined, they suggest a coarse-to-fine workflow: radar screening, microwave confirmation of extent, and ultrasonic detail.

D. Robot Platform

Carbon-fiber composites can reduce weight and energy use while improving load capacity and impact resistance. Gecko-inspired adhesive feet with controllable adhesion could complement rotor thrust on vertical surfaces. Fusing LiDAR, vision, ultrasonic, and infrared sensing would let the robot build a 3D map of its surroundings. Combining reinforcement learning with rule-based decision trees would then support autonomous obstacle avoidance and path planning.

E. Management Platform

A unified tunnel quality database spanning railway, highway, and metro tunnels under diverse geology would enable deeper big-data analysis and AI-assisted decision support for remediation planning.

XI. CONCLUSION

We presented an automated tunnel lining inspection system that couples a wall-climbing robot carrying air-coupled GPR with a lightweight signal network and a 3D management platform. The robot removes the cable tether and the need for lift trucks or scaffolds. Working directly on raw radar traces, the network classifies four defect-related signal types with a macro-F1 of 0.93 and estimates void depth, and it generalizes better than an image-based YOLOv5s baseline. The platform turns each detected defect into a traceable, closed-loop record. Field deployment over more than 160 km of lining shows that the approach can replace manual high-altitude surveys and shorten the cycle from detection to remediation.

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